WILLIAMS V.S. TUBMAN UNIVERSITY

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College of Engineering and Technology

Department of Computer Science

PROJECT PLAN

Execution Roadmap & Schedule

Version 1.0 | March 2026

Prepared for: Computer Lab Management Team, Department of Computer Science

Project Team: Software Engineering Group 2 (6 Members)

Academic Supervisor: Prof. Kwitee D. Gaylah

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1. EXECUTIVE SUMMARY

This Project Plan outlines the execution strategy for the Tubman University Computer Lab Inventory System. Developed as part of the Software Engineering course curriculum, this project aims to deliver a functional desktop application that manages equipment assets, borrowing workflows, and maintenance records for the Department of Computer Science laboratories.

This document aligns with the approved Software Requirements Specification (SRS) and adheres to the academic timeline provided by the course instructor. It details the phases of development, team responsibilities, risk mitigation, and quality assurance strategies required to successfully complete the project by the final presentation date.

2. PROJECT SCOPE & OBJECTIVES

2.1 Project Objectives

- **Develop a Functional Desktop Application: Build an offline-first inventory system using Electron, Nest.js, and SQLite.**
- **Apply Engineering Principles: Demonstrate SOLID principles, Agile methodology, and version control best practices.**
- **Meet Academic Deadlines: Complete all phases (Requirement, Design, Development, Testing, Presentation) within the scheduled timeline.**
- **Deliver Value to Department: Provide the Computer Lab with a tool to track equipment and reduce loss.**
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2.2 Scope Boundaries

- **In-Scope: Equipment catalog, checkout/check-in workflow, maintenance tracking, alerts, reporting, user authentication, offline backup.**
- **Out-of-Scope: Cloud synchronization, mobile app version, integration with university ERP, payment processing.**

3. PROJECT TIMELINE & PHASES

The project follows the academic schedule provided by the Software Engineering course. The timeline is compressed, requiring focused effort during development and testing phases.

Duration	Phase	Milestone Deliverables	Status
March 30-31	**Requirement Gathering & SRS Document**	**- Signed SRS Document - Stakeholder Approval - Initial Backlog**	✅ **Completed**
April 1	**Designs (SDD)**	**- Sequence Diagram - Wireframes - Use Case Diagram - Activity Diagram - ER Diagram - Component Diagram - Deployment Diagram**	🔄 **In Progress**
April 15-22	**Development/Agile & Iterative Module**	**- Functional Codebase - Core Features Implemented - Daily Standup Logs - Sprint Backlog**	⏳ **Pending**
April 29-May 4	**Testing/Quality**	**- Unit Testing Reports - System Testing Reports - Bug Fix Log - Test Coverage Metrics**	⏳ **Pending**
May 5	**Feedback**	**- User Feedback Form (Lab Technician) - Faculty Review Comments - Change Request Log**	⏳ **Pending**
May 11	**Final Presentation of Final Project**	**- Final Project Demo - Source Code Repository - Final Report - Presentation Slides**	⏳ **Pending**

3.1 Phase Descriptions

Phase 1: Requirement Gathering (March 30-31)

- **Activity: Finalize SRS, define user stories, get supervisor sign-off.**
- **Owner: Project Lead.**
- **Output: Approved SRS Document.**

Phase 2: System Design (April 1)

- **Activity: Create all UML diagrams, design database schema, create UI wireframes.**
- **Owner: Database Developer (ERD), Frontend (Wireframes), Lead (Use Case/Activity).**
- **Output: Software Design Document (SDD) with all diagrams.**

Phase 3: Development (April 15-22)

- **Activity: Core coding sprints. Implementing Auth, Equipment, Checkout, and Reports.**
- **Owner: Entire Team (Frontend, Backend, Database, DevOps).**
- **Output: Working software prototype (Beta).**

Phase 4: Testing (April 29-May 4)

- **Activity: Unit tests, integration tests, system testing, bug fixing.**
- **Owner: QA Engineer + Entire Team.**
- **Output: Test Reports, Stable Release Candidate.**

Phase 5: Feedback (May 5)

- **Activity: Demo to Lab Technician and Faculty; collect feedback.**
- **Owner: Project Lead.**
- **Output: Feedback Report.**

Phase 6: Final Presentation (May 11)

- **Activity: Final demo, code submission, report submission.**
- **Owner: Entire Team.**
- **Output: Final Grade Submission.**

4. TEAM ROLES & RESPONSIBILITIES

The team consists of 6 members, each with specific responsibilities aligned with the project phases.

Role	Member	Primary Responsibilities	Key Deliverables
Project Lead / Scrum Master	**Princeton**	**Plan coordination, SRS/SDD oversight, stakeholder communication, code review**	**Project Plan, SRS, Final Report, Meeting Notes**
Frontend Developer	**Regestopher**	**UI implementation, Wireframes, React components, Electron window**	**Wireframes, UI Components, Desktop Window**
Backend Developer	**Philip**	**API design, Business logic, Authentication, Sequence Diagrams**	**API Code, Sequence Diagrams, Auth Module**
Database Developer	**Oliver**	**DB Schema, ER Diagram, TypeORM entities, Data integrity**	**ER Diagram, Database Schema, Migrations**
QA/Testing Engineer	**Ernest**	**Test plans, Unit/System testing, Bug tracking, Activity Diagrams**	**Test Reports, Bug Log, Activity Diagrams**
DevOps/Build Engineer	**Alice**	**Deployment Diagram, Component Diagram, Build scripts, Installer**	**Deployment Diagram, .exe Installer, CI/CD**

5. DEVELOPMENT APPROACH (AGILE & ITERATIVE)

Although the academic timeline presents distinct phases, the development phase (April 15-22) will utilize Agile Scrum principles to ensure flexibility and quality.

5.1 Sprint Structure (Within Development Phase)

- **Sprint 1 (April 15-18): Core Infrastructure (Auth, Database, Basic UI).**
- **Sprint 2 (April 19-22): Feature Implementation (Equipment, Checkout, Reports).**

5.2 Daily Standups

- **Time: Every day during Development phase (April 15-22).**
- **Format: 15 minutes (WhatsApp Voice Note or Quick Call).**
- **Questions: What did I do yesterday? What will I do today? Any blockers?**

5.3 Version Control Workflow

- **Repository: GitHub (Public).**
- **Branching: main (stable), develop (integration), feature/xyz (tasks).**
- **Pull Requests: All code must be reviewed before merging to develop.**

6. DELIVERABLES & MILESTONES

Each phase in the timeline corresponds to specific tangible deliverables.

Phase	Deliverable	Format	Owner
Requirement	**SRS Document**	**PDF/Word**	**Project Lead**
Design	**SDD Document (Diagrams)**	**PDF/Word**	**All Members**
Development	**Source Code**	**GitHub Repo**	**All Members**
Development	**Desktop Installer (.exe)**	**Executable File**	**DevOps**
Testing	**Test Report**	**PDF**	**QA Engineer**
Feedback	**Feedback Summary**	**Word/PDF**	**Project Lead**
Presentation	**Final Slides & Demo**	**PPT + Live Demo**	**All Members**

7. RISK MANAGEMENT PLAN

Given the compressed timeline, risks are monitored closely.

Risk	Likelihood	Impact	Mitigation Strategy	Owner
Timeline Slip	**High**	**High**	**Prioritize P1 (Critical) features; defer P3 (Nice-to-have) if behind schedule.**	**Project Lead**
Technical Blockers	**Medium**	**High**	**Early spike tasks (April 1-14); Use proven libraries (Nest.js, TypeORM).**	**Backend/DevOps**
Team Availability	**Medium**	**High**	**Clear task assignment; Daily standups to catch delays early.**	**Project Lead**
Bug Accumulation	**High**	**Medium**	**Continuous testing during development; dedicated Testing phase (April 29-May 4).**	**QA Engineer**
Hardware Failure	**Low**	**High**	**Daily GitHub commits; Cloud backup of code; Local DB backups.**	**DevOps**

8. QUALITY ASSURANCE & TESTING STRATEGY

Testing is integrated throughout but emphasized during the April 29-May 4 phase.

8.1 Testing Levels

1. **Unit Testing: Testing individual functions (Backend Services, Frontend Components).**
2. **Integration Testing: Testing API endpoints with Database.**
3. **System Testing: Testing the full desktop application workflow.**
4. **User Acceptance Testing (UAT): Feedback session on May 5 with Lab Technician.**

8.2 Quality Standards

- **Code Style: ESLint + Prettier enforced.**
- **Documentation: All functions commented (JSDoc).**
- **Testing: Critical paths must have unit tests.**
- **Security: Passwords hashed (bcrypt); SQL injection prevented (TypeORM).**

9. COMMUNICATION PLAN

Effective communication ensures the team stays aligned within the short timeline.

Channel	Purpose	Frequency	Participants
WhatsApp Group	**Daily updates, quick questions, blockers**	**Daily**	**Entire Team**
GitHub Issues	**Task tracking, bug reports, feature requests**	**As needed**	**Entire Team**
Email	**Formal submission to Supervisor, documentation**	**Milestone Based**	**Project Lead + Supervisor**
In-Person/Virtual	**Design reviews, Final Presentation prep**	**April 1, May 5, May 11**	**Entire Team + Supervisor**

10. APPROVAL & SIGN-OFF

By signing below, team members confirm they understand the timeline, their roles, and the deliverables required for successful project completion.

Name	Role	Signature	Date
	Project Lead / Student Developer		
	Frontend Developer		
	Backend Developer		
	Database Developer		
	QA/Testing Engineer		
	DevOps/Build Engineer		
	Academic Supervisor		

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